

The Verification Bridge

Consolidating the Multi-Track Substrate Expansion into a Per-Theorem Verified Index:

the Honest Hinge from Expansion to Verified Anatomy

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Doctrine v11 LOCKED — 749 declarations · 14 unique axioms · 163 sorries @ c7c0ba17 · Λ is **Conjecture 1** · SLSA L1+L2 (not L3)

Abstract

Between the multi-track *substrate expansion* (v18, 2026-05-30) and the formally-verified *anatomical substrate* (v20, 2026-06-01), the SZL Holdings Ouroboros line passed through a short but pivotal phase whose product was not new breadth but new *discipline*: the consolidation of a rapidly expanding Lean 4 corpus into a per-theorem verified index with an honest, machine-checkable status for every claim. This paper, v19 “The Verification Bridge,” documents that phase. v18 added roughly twenty-nine modules across coding theory, differential privacy, PAC-Bayes generalization, mechanism design, certified robustness, and a Putnam-style hardening track; many landed with *open obligations* (sorry placeholders) recorded honestly rather than papered over. The bridge work asked a single governance question of that corpus: *which results are actually kernel-verified, which are sorry-free only in fragments, which depend on declared idealizing axioms, and which remain open?* We answer it with a verbatim per-theorem Lean ledger, reporting exact sorry/axiom/theorem counts per file. The central methodological commitment is the **honesty doctrine** inherited from v14 and carried verbatim to v23: the Lutar invariant $\Lambda_k(x) = (\prod_i x_i)^{1/k}$ is **Conjecture 1** unconditionally — never a theorem; the locked kernel proves exactly five formulas; SLSA is L1+L2, not L3; and every idealizing axiom is disclosed in a #print axioms ledger. v19’s contribution is the *bridge itself*: the move from “we shipped many modules” to “here is exactly what each module proves, with a per-theorem status ledger,” which is precisely the externalized meta-level the v20 anatomy and the v23 unification build upon.

Keywords: formal verification, Lean 4, per-theorem index, governance substrate, honesty doctrine, sorry accounting, drift gate, differential privacy, PAC-Bayes, mechanism design, certified robustness, Reed–Solomon, supply-chain attestation.

*Honesty note (verbatim): The Lutar invariant Λ is **Conjecture 1** unconditionally and is never claimed proven unconditionally. Exactly five formulas are locked/proven at c7c0ba17. Modules that carry sorry obligations are reported as carrying them. Declared idealizations are disclosed. SLSA L1+L2 attested, not L3.*

1. Introduction: why a bridge, and why it was almost skipped

The Ouroboros thesis line records the intellectual provenance of the SZL governance substrate as a sequence of versioned, DOI-pinned papers. For a period the lineage read “v18 → v20” with a documented gap at v19: the numbering jumped during a late-May 2026 consolidation, and the gap was annotated as intentional rather than as a missing artifact. This paper fills that gap *honestly*. It does not invent a result that did not happen; it documents the work that did happen between v18 and v20 and that was, at the time, folded silently into the v20 consolidation instead of being written up as its own version.

That work has a clear and defensible identity. v18 (“Multi-Track Substrate Expansion”) was an *expansion* release: it grew the Lean corpus across many mathematical tracks at once. v20 (“The Culmination”) was a *verified-anatomy* release: it presented the substrate as a formally-verified anatomical system. Between the two sits the act that makes the second possible from the first — **verification consolidation**: taking a fast-growing, partly-open corpus and imposing a per-theorem index in which every claim carries an explicit, machine-checkable status. That act is the bridge.

Thesis of v19. A governance substrate earns trust not by the *number* of modules it ships but by the *legibility of their status*. The right unit of accounting is the *theorem*, not the module or the paper: for each Lean declaration the substrate must be able to state, mechanically, whether it is kernel-verified with only Lean-core axioms, sorry-free in a named fragment, sorry-free only under a declared idealizing axiom, or still open. We construct that per-theorem index over the v18 corpus and the 2026-05-30 expansion track, and we show that the resulting honesty ledger — not any single new theorem — is the load-bearing contribution that the v20 anatomy and the v23 unification inherit.

What v19 is not. v19 is not a claim of new proven mathematics. The five locked formulas were locked before it; the conditional Λ -uniqueness result and the machine-checked refutation of unconditional uniqueness are later (Wave-2 through Wave-4) work reported in v22 and v23. v19’s honest scope is *consolidation and accounting*: it standardizes how the substrate reports proof status, and it makes the v18 expansion auditable. We state this boundary plainly because the honesty doctrine forbids rounding a bookkeeping contribution up to a mathematical one.

2. Related work: verified kernels and verifiable claims

The bridge sits in a well-developed tradition of *machine-checked correctness* and a younger tradition of *verifiable claims for AI governance*; v19 borrows the discipline of the former and serves the goals of the latter.

2.1 Kernel-checked systems and proof assistants

The gold standard for trust-by-checking is a small, audited proof kernel that mechanically certifies every step. The Lean 4 theorem prover [de Moura & Ullrich 2021] and its mathematical library Mathlib [mathlib Community 2020] are the substrate’s formal foundation: a Lean development is *sorry-free* when no placeholder remains and the kernel checks the whole term, and its trusted base is exposed by the `#print axioms` command. The seL4 verified operating-system kernel [Klein et al. 2009] and the CompCert verified C compiler [Leroy 2009] established the cultural expectation we adopt: a verified artifact states *exactly* what is proven and under which assumptions, and treats every undischarged obligation as a first-class, named object rather than a silent gap.

2.2 Verifiable claims for AI governance

The complementary line is the argument that AI developers should be able to make *verifiable claims* — claims an external party can check — rather than asking to be trusted [Brundage et al. 2020]; national policy bodies have echoed the demand that developers be able to *prove* properties of their systems [NTIA 2024]. The bridge is the operational answer to a precondition of that program: before you can make a verifiable claim about a corpus, you must have a mechanical account of *which* claims are checkable and *to what degree*. The per-theorem index is that account.

2.3 The disclosed-axiom discipline

Where a proof must assume an idealization (e.g. cryptographic collision resistance), the honest practice — followed by the transparency-protocol verification literature [ProVerif 2023] and by certificate-transparency design [RFC 6962] — is to name the assumption and expose it. v19 standardizes this as a column in the index: an axiom-gated result is sorry-free *given* a disclosed axiom, and is never conflated with an unconditional one.

3. Formal preliminaries: definitions, the axiom system, and notation

We fix the vocabulary the index uses. The objects are Lean 4 declarations; the predicates are properties of their dependency on the trusted base.

3.1 Proof-status predicates

For a Lean declaration d we define four mutually informative status predicates, each mechanically decidable from a build:

- **kernel-verified / sorry-free (Lean-core):** d elaborates with no sorry, and `#print axioms d` returns a subset of the Lean-core trio {`propext`, `Quot.sound`, `Classical.choice`}. This is the strongest status.
- **sorry-free fragment:** a named sub-result of a module is sorry-free even though the enclosing module still carries open obligations elsewhere. The fragment’s status is asserted only for the named declaration.
- **axiom-gated:** d is sorry-free, but `#print axioms d` lists one or more *declared* idealizing axioms (e.g. a hash collision-resistance axiom). The result is true *conditional on* the disclosed axiom and is labeled as such.
- **open:** d contains at least one sorry, recorded as a typed, named obligation with a discharge route. Open is reported as open.

3.2 The Lutar invariant and its axioms

The trust aggregator under study is the **Lutar invariant**, the equal-weight geometric mean

$$\Lambda_k(x_1, \dots, x_k) = (\prod_{i=1}^k x_i)^{1/k}, \text{ defined on } [0,1]^k,$$

intended to combine multi-axis trust sub-scores into a single governance scalar. The candidate axioms a governed aggregator might satisfy — carried verbatim into v22/v23 — are:

Axiom	Name	Statement (informal)
A1	Monotonicity	Φ is non-decreasing in each argument.

Axiom	Name	Statement (informal)
A2	Positive homogeneity	$\Phi(rx) = r \cdot \Phi(x)$ for $r > 0$ (ratio-scale invariance).
A3	Idempotence	$\Phi(c, \dots, c) = c$ on the diagonal.
A4	Boundedness / normalization	$\min x_i \leq \Phi(x) \leq \max x_i$; fixes endpoints.
A5	Permutation invariance	Φ is symmetric under any reordering of its arguments.
A6'	Block-consistency	Aggregating block means via Φ equals aggregating all elements (declared, governance-natural).

v19 carries only the **label** on $\wedge$ (Conjecture 1) and the definitions; the conditional uniqueness theorem under A6' and the machine-checked refutation of unconditional uniqueness under {A1–A5} are later results (§6). The notation is fixed here so the later versions inherit it.

4. The v18 baseline and the per-theorem verified index

4.1 What the expansion shipped

v18 expanded the corpus along multiple independent tracks. To make the bridge concrete we first fix the v18 baseline as it stood at the substrate-expansion release, drawing per-file counts from the corpus index (`lean_per_version.json`) rather than from prose. The v18 thesis-theorem track (`Lutar/Thesis/TH_V18_*`) is sixteen modules first committed 2026-05-29; the majority are sorry-free with only Lean-core axioms, a minority carry a single honest open obligation, and `TH_V18_14_SHA256CollisionHonest` is sorry-free but *axiom-gated* (two declared collision-resistance axioms). Two further governance/provenance modules belong to the window: `Lutar/Gates/Adinkra` (sorry-free, one declared axiom, three theorems) and `Lutar/SBOMPProvenance` (three sorries, two declared axioms, six theorems), the formal counterpart of the SLSA L1+L2 posture.

4.2 The bridge artifact: a per-file (sorry, axiom, theorem) ledger

The bridge artifact is a per-file ledger for the 2026-05-30 expansion track (the “v18–v19 window”), read verbatim from the corpus index. We report it honestly: many modules are *not* fully closed, and the index says so. The point is not to claim them closed but to make their open status auditable at the theorem level.

Module (Lutar/...)	sorry	ax	thm	Status
CodingTheory/ReedSolomonSingleton	7	0	8	open (fragments)
DP/GaussianMechanism	5	0	8	open (fragments)
DP/RDPCComposition	1	0	6	open (1 obligation)
PACBayes/PACBayes	7	1	10	open, 1 declared axiom
PACBayes/CapabilityImprovementRate	5	0	7	open (fragments)
MechanismDesign/VCG	9	0	5	open (later proven on-branch, v22)
Robustness/CertifiedRadius	4	0	5	open (fragments)
Putnam/P_A1	1	0	5	near-closed
Putnam/P_A2	8	0	7	open (fragments)
Putnam/P_A3	0	0	4	sorry-free

Module (Lutar/...)	sorry	ax	thm	Status
Putnam/P_A4	4	0	3	open (fragments)
Putnam/P_A5	6	0	3	open (fragments)
Putnam/P_A6	2	0	11	open (fragments)
Putnam/P_B1 – P_B6	30	0	20	open (fragments)

The ledger is deliberately unflattering. Of the nineteen expansion-track modules, exactly one (Putnam/P_A3) is fully sorry-free; the rest carry open obligations counted to the theorem. This is the honest state of the v18 expansion at the moment of the bridge, and recording it *as such* is the contribution. The later closure of several of these (e.g. VCG truthfulness proven on-branch, reported in v22) is downstream work; v19 records the starting line, not a finish it did not reach.

Remark (why open obligations are kept, not hidden). A naive consolidation would delete or comment out incomplete proofs to present a clean ‘green’ corpus. The bridge refuses this. Every sorry is a typed, named open obligation with a discharge route; deleting it would convert a disclosed gap into a silent one. The honesty doctrine treats a disclosed obligation as strictly better than a hidden assumption, in the Lakatosian sense of explicit, revisable lemma-conditions [Lakatos 1976].

5. The kernel-hardening invariant: locked vs. experimental scope

The bridge’s organizing principle is a strict separation between a *locked kernel* and an *experimental scope*, enforced mechanically.

5.1 The locked kernel and the five locked formulas

The Lean formalization has a locked kernel pinned at commit c7c0ba17 (Doctrine v11): **749 declarations / 14 unique axioms / 163 sorries** (counted as 112 baseline + 51 Putnam). Within this counted scope, exactly **five** PURIQ governance formulas are proven and locked. We disclose each with its #print axioms footprint:

Formula	Statement (Lean core)	#print axioms	Status
F1 replay determinism	$f(x) = f(x); xs.map\ f = xs.map\ f$	(none)	LOCKED
F11 Ayni reciprocity	$(b + c) - c = b$ over $\mathbb{Z}$	[propext]	LOCKED
F12 Kuramoto additive	$k \cdot (p_1 + p_2) = k \cdot p_1 + k \cdot p_2$ over $\mathbb{N}$	(none)	LOCKED
F18 Reed–Solomon RS(10,6)	$10 - 6 = 4; e \leq 4 \Rightarrow 10 - e \geq 6$	[propext, Quot.sound]	LOCKED
F19 Bekenstein additive	$s_1 \leq s_1 + s_2$ (additive scaffolding only)	(none)	LOCKED

F12 and F19 prove only the *additive* fragments of, respectively, Kuramoto synchronization and the Bekenstein bound; they are honest scaffolding, never described as the full physical theorems. This disclosure is itself part of the bridge: the index records not just “F12 proven” but “F12 proves the additive fragment, with no axioms.”

5.2 The drift gate

Everything in the 2026-05-30 expansion track (§4.2) lives in an *experimental* scope counter-excluded from the locked count and excluded until re-audited under the authoritative lake build. The kernel

hardening v19 standardizes is the **drift gate**: a CI check that the locked declaration/axiom/sorry counts do not move when experimental modules are added. This is the bridge made executable — the guarantee that “we added modules” can never silently become “we changed what is proven.”

Proposition 5.1 (Scope-separation invariant, informal) [open obligation]

Adding an experimental module changes the *live* corpus counts but must leave the *locked* counts (749, 14, 163) at c7c0ba17 invariant. A change to the locked counts is a doctrine violation and fails CI.

Proof / Lean reference. *This is not a theorem about mathematics; it is a theorem about bookkeeping discipline, enforced by a CI comparison against a pinned baseline. It is exactly the externalized meta-level (Lean kernel $\rightarrow$ CI $\rightarrow$ human sign-off) that v23 later articulates in full.*

5.3 A worked drift-gate trace

To make the gate concrete, consider the CI sequence when the agentic-loop module (§6) was first pushed. The kernel check and the doctrine check passed immediately, but the drift gate *failed* — precisely because the new file’s declarations had not yet been excluded from the locked baseline, so the live count no longer matched the pinned 749/14/163. The doctrine-correct repair was not to weaken the gate but to register the new namespace in EXPERIMENTAL_SCOPES, after which the gate went green with the locked counts untouched. This is the gate behaving exactly as designed: a new module cannot enter without an explicit, reviewed decision about whether it is locked or experimental, and the locked numbers are never silently moved.

- 1 **Push.** A new experimental module is added; the live declaration count rises.
- 2 **Kernel check + doctrine check: green.** The mathematics still type-checks; the locked formulas are untouched.
- 3 **Drift gate: red.** The live count differs from the pinned baseline, because the new declarations are not yet scoped.
- 4 **Repair.** Register the namespace in EXPERIMENTAL_SCOPES — an auditable, reviewed scoping decision, not a count edit.
- 5 **Re-run: green.** Live build passes, locked counts (749/14/163) verified invariant, experimental theorems tracked separately.

The trace shows why the gate is the load-bearing artifact of the bridge: it converts the abstract claim “adding modules must not change what is proven” into a red/green signal that a reviewer cannot route around without leaving a record.

6. The index format under load: end-to-end verification of the agentic loop

The bridge’s thesis — that the right unit of accounting is the theorem with a machine-checkable status — is only convincing if the format survives contact with the substrate’s hardest claim: that the agentic control loop itself (retrieve $\rightarrow$ plan $\rightarrow$ tool-call $\rightarrow$ policy-check $\rightarrow$ kernel-check $\rightarrow$ emit) is not merely architecturally sound but *formally verified as a system*. We therefore stress-test the index format against a real, CI-green, experimental-scope development that models the loop as a small-step operational system in Lean 4 and proves six end-to-end properties over whole runs. The point here is not to claim these theorems are part of the *locked* kernel — they are explicitly experimental scope,

excluded from the locked 749/14/163 count — but to show that the per-theorem index assigns each one an honest status, with its #print axioms footprint, exactly as the format demands.

6.1 The model: hops, receipts, and a quarantined untrusted channel

The loop is modeled Mathlib-free as a six-constructor hop type and an append-only receipt chain in which each executed hop emits exactly one content-addressed receipt linked to its predecessor by a hash. The state carries an *untrusted* retrieved payload (the adversary-controlled RAG context, treated as a HIGH-security input), two gate verdicts (policy and kernel, the LOW gate inputs), the running predecessor hash, and the receipt log. The load-bearing modeling decision is that gate verdicts are computed from the gate inputs *only* and are never written from the untrusted blob: the retrieve hop *records* the untrusted payload in a receipt but *quarantines* it from the decision.

```
inductive Hop | Retrieve | Plan | ToolCall | PolicyCheck | KernelCheck | Emit
inductive Decision | Allow | Deny
opaque hashFn : Nat -> Nat -- abstract content hash
structure Receipt where hop : Hop; payload : Nat; decision : Decision
prevHash : Nat; selfHash : Nat
def step (s : St) (h : Hop) : St -- append one chained receipt
def run (s0 : St) (prog : List Hop) : St := prog.foldl step s0
def loopProgram : List Hop :=
[Retrieve, Plan, ToolCall, PolicyCheck, KernelCheck, Emit]
```

6.2 Six end-to-end properties, each with a disclosed axiom footprint

The index records all six properties (28 kernel-verified theorems, sorry-free, exit 0, no sorryAx) with their verbatim #print axioms status. Fourteen are fully axiom-free, ten use only the Lean-core trio, and exactly four are axiom-gated on a single disclosed crypto idealization. This is the index format applied to the substrate’s headline architectural claim:

Property	What is proven over whole runs	#print axioms	Status
P1 receipt-completeness	every hop appends exactly one chained receipt; log append-only; chain contiguous; full loop $\Rightarrow$ exactly 6 receipts	[propext, Quot.sound]	sorry-free
P2 gate-soundness	Emit is ALLOW iff both policy and kernel gates ALLOW; a single DENY is absorbing	[propext]	sorry-free
P3 non-interference	mutating the untrusted RAG payload provably cannot flip a DENY into an ALLOW (Goguen-Meseguer)	(none) for p3a/b/c	axiom-free
P4 replay-determinism	replaying a run from the same state yields a byte-identical receipt chain (F1 lifted to whole loop)	(none)	axiom-free
P5 tamper-evidence	any single-receipt payload mutation is rejected on chain re-verification	[..., hashFn_collision_resistant]	axiom-gated
P6 monotone auditability	the verifier’s accept set is prefix-closed: a valid prefix stays valid as the log grows	[propext]	sorry-free

The single declared axiom, used only by P5, is the standard collision-resistance idealization


```
axiom hashFn_collision_resistant :  $\forall a b : \mathbb{N}, \text{hashFn } a = \text{hashFn } b \rightarrow a = b$ 
```

disclosed exactly as the F13', TH_V18_14, and Merkle-tree theorems disclose it (NIST FIPS 180-4; [Merkle 1987]). The non-interference result is the one the index most wants to surface honestly: P3 is the formal content of "prompt-injection or poisoned retrieval cannot flip a verdict," and its core is *axiom-free*. The companion non-vacuity lemma proves the quarantine is not hollow — the untrusted blob genuinely flows into the audit log, yet provably never into the verdict, the Goguen–Meseguer information-flow condition [Goguen & Meseguer 1982] applied to the whole loop.

Scope note (honest). These 28 theorems are *experimental scope* in a separate namespace, NOT imported into the locked library and excluded from the drift baseline; the locked counts 749/14/163 at c7c0ba17 and locked_proven = 5 are UNCHANGED. The bridge claims only that its index assigns each theorem an honest, machine-checkable status — not that the loop verification is part of the locked kernel.

7. Experimental theorem waves: the index format at corpus scale

Beyond the agentic loop, the bridge format is exercised against two larger experimental theorem waves that the substrate later accrued. We report them here as the index would: with honest maturity labels, the declared-axiom footprint, and an explicit statement of what could *not* be closed. None of these touch the locked kernel.

7.1 Wave-5: inequalities, quantum bounds, and anti-deflation martingales

The wave-5 track adds eleven theorems spanning ratio-scale inequalities and stochastic audit bounds, all CI-green at their pinned Mathlib. The most relevant to the Λ line is the AM–GM no-inflation result, which proves the geometric mean never exceeds the arithmetic mean — the formal statement that Λ *cannot inflate* a trust score above the naive average:

Theorem	Statement	Axiom footprint
W5-1 AM–GM no-inflation	$\Lambda(x) \leq \text{arithmetic mean (geometric mean never inflates)}$	Mathlib re-export
W5-1b two-point AM–GM	$\sqrt{ab} \leq (a+b)/2$	Lean-core
W5-2 Cauchy–Schwarz	$(x \cdot y)^2 \leq (x \cdot x)(y \cdot y)$ (inner-product form)	Mathlib re-export
C1 Tsirelson bound	quantum CHSH value $\leq 2\sqrt{2}$	[propext, Classical.choice, Quot.sound]
C2 CHSH classical bound	classical CHSH value ≤ 2	[propext, Classical.choice, Quot.sound]
C6 Jensen forecaster	convexity bound for a calibrated forecaster	[propext, Classical.choice, Quot.sound]
W5-5 optional-stopping	anti-deflation: a fair audit martingale is not driven down by stopping	(zero axioms)

The wave honestly records what is *not* achievable at the pinned Mathlib: the Hoeffding (C3), Azuma (C4), and KL-non-negativity (C5) targets returned HTTP 404 against the pinned library revision and are reported as **not achievable** rather than faked. At its head the experimental wave-5 corpus stood at 1189 declarations / 19 unique axioms / 308 sorries — numbers an order of magnitude looser than the

locked kernel, which is precisely why they are kept in experimental scope. The locked 749/14/163 at c7c0ba17 and `locked_proven = 5` are unchanged.

7.2 Wave-7: invariance, rank-count, PAC-Bayes, and a two-sided audit envelope

The wave-7 track adds ten theorems: degree-sum isomorphism-invariance (W7-1/1a), conformal rank-count laws (W7-4a/b/c), the PAC-Bayes ordering $\min \leq \text{avg} \leq \max$ (W7-5/5a/5b), and a Doob two-sided audit envelope (W7-6/6a). As in wave-5, the Hoeffding/Azuma/KL targets remain HTTP 404 at the pin and are reported unachieved. The PAC-Bayes ordering is the load-bearing one for governance: it bounds the realized generalization risk of a governed policy between the best- and worst-case posterior members, a measured-not-proven operating guarantee the runtime can cite [McAllester 1999]. The two-sided Doob envelope gives an auditor a martingale band within which a fair process stays with high probability — the stochastic analogue of the deterministic drift gate.

Why these are reported, and reported as experimental. A dishonest consolidation would fold these thirty-nine experimental theorems into a headline ‘the substrate proves 90+ theorems’ count. The bridge refuses. Each is labeled with its maturity (experimental scope), its axiom footprint, and its CI status; the locked kernel of five formulas is reported separately and never inflated. The index format scales precisely because it never collapses these distinctions.

8. The Λ honesty section: Conjecture 1, conditional, and the counterexample

Although v19 itself proves no Λ result, it is the version that *fixes the label Λ* carries through the rest of the line, so we state the full epistemic structure that v22/v23 later certify. This section is carried forward verbatim and cites the two governing internal references: the Λ -axiom defense and the geometric-mean forcing analysis.

8.1 Three propositions, three statuses

A philosophically careful reading distinguishes three propositions, each with a different epistemic status. The substrate asserts exactly the truth value of each, and never lets a reader collapse them:

Proposition	Statement	Status	Evidence
(U) Unconditional uniqueness	Λ is the unique aggregator satisfying {A1–A5}.	FALSE	Machine-checked counterexample <code>maxAgg_ne_Lambda</code> , CI-green.
(C) Conditional uniqueness	Given {A1–A5}+A6′, Λ is unique.	PROVEN (cond.)	CI-green, kernel-checked; A6′ disclosed in #print axioms.
(Conj) Governance conjecture	The right axiomatization of a governed aggregator forces Λ .	Conjecture 1	Open philosophical question whether A6′ is the correct demand.

8.2 The counterexample that makes the labeling honest

Unconditional Λ -uniqueness under {A1–A5} is machine-checked **false**: the max-aggregator satisfies A1–A5 yet differs from Λ . The witness is sorry-free and CI-green; concretely at the point (4,1) the

max-aggregator returns $\max=4$ while $\Lambda_2(4,1) = \sqrt{4} = 2$, so they disagree. Equivalently the asymmetric mean $\Phi(x_1, x_2) = x_1^{2/3} \cdot x_2^{1/3}$ satisfies A1–A4 but fails permutation invariance (A5), so A1–A4 alone do not even force symmetry, let alone Λ . This refutation is the *maximal-true-statement* discipline of [Lakatos 1976]: the monster is exhibited, the guilty lemma (A6') is named, and the revised conditional theorem supersedes the primitive conjecture.

8.3 Why honest labeling is a virtue, not a weakness

The internal Λ -axiom defense argues, on grounds from formal epistemology, that the calibrated three-tier labeling (locked theorem / conditional theorem / conjecture) is what *makes the positive claims credible*. By process reliabilism [Goldman 1979], a label is trustworthy iff the labeling process has a high accuracy rate; a system that calls everything “proved” carries less information in its “proved” label than one that also accurately says “conjecture.” By Nozick’s sensitivity condition, the counterexample is the mechanism that causes the conjecture label: were unconditional uniqueness true, the system would not so label it. The geometric-mean choice itself is independently motivated by measurement theory — trust sub-scores are ratio-scale, and the geometric mean is the meaningful central tendency for ratio data [Stevens 1946]; [Narens 2002] — and the geometric mean is the unique quasi-arithmetic mean satisfying reciprocity and positive homogeneity [Aczél & Saaty 1983], with A6'-style block-consistency the published, governance-natural route [Csátó 2018].

Conjecture 1 (The Lutar invariant; never a theorem). The equal-weight geometric mean Λ_k is the correct unique trust aggregator for governed AI. This is an open claim about the *right* axiomatization, not a mathematical theorem; the substrate carries the ‘Conjecture 1’ label on Λ in every artifact, including this one.

9. Architecture of the index: the externalized meta-level

The bridge is not only a document; it is an executable pipeline. We describe its three-layer architecture, the same meta-level the v20 anatomy and the v23 unification later name explicitly.

9.1 Layer 1 – the corpus index

A machine-readable index (`lean_per_version.json`) records, per file and per declaration, the version window, the sorry count, the declared-axiom count, and the theorem/lemma count. It is harvested from the source tree with disclosed limits (it is an index, not the authoritative kernel count).

9.2 Layer 2 – the authoritative kernel build

The authoritative numbers come from a full `lake build` of the locked kernel at `c7c0ba17`, which the kernel checks end to end. The locked counts (749/14/163) are defined by this build, not by the index. The drift gate compares the live build against the pinned baseline on every CI run.

9.3 Layer 3 – human sign-off

A release is gated on a human sign-off that the disclosed status matches the build. The chain — Lean kernel → CI drift gate → human sign-off — is the substrate’s trusted computing base for *claims about the corpus*, distinct from (and layered above) the trusted base for the mathematics itself.

10. Empirical operating characteristics (measured, not proven)

The bridge discipline is cheap to run; we report measured operating characteristics from the substrate, each labeled **measured-not-proven**. These are reproducible operational facts, not theorems.

Quantity	Measured value	Conditions
Drift-gate check	sub-second per CI run	compares live build counts vs pinned 749/14/163
Receipt build (p50 / p99)	11.5 μ s / 50.7 μ s	content-hashed receipt emission
Receipt verify	10.4 μ s	hash-chain link check
Λ_9 aggregate	3.12 / 3.29 μ s	9-axis geometric-mean evaluation
Governance overhead (p50 / p99)	0.49–0.59 ms / 1.27 ms	over 24,800 governed calls
p-closure	100% (8,000 / 8,000)	audit-closure pass rate
Replay determinism	5x byte-identical	repeated replay of a recorded trace

These figures characterize the runtime the index governs; they do not bear on what is proven. The honesty doctrine keeps the measured column strictly separate from the proven column — a measured latency is never described as a theorem, and a theorem is never described as a benchmark.

11. Position in the lineage

The bridge sits at a specific and now-documented place in the arc.

Ver	Date	Role relative to the bridge
v18	2026-05-30	Expansion. 29 modules across many tracks; per-theorem index begun; many open obligations recorded.
v19	2026-05-31	The Verification Bridge (this paper). Consolidation into a per-theorem verified index; locked-vs-experimental scope separation and the drift gate; honesty doctrine as acceptance criterion.
v20	2026-06-01	Culmination. The formally-verified anatomical substrate, presentable because the bridge made per-theorem status auditable.
v21	2026-06-01	PURIQ-OS substrate: the 12-organ runtime that executes the verified anatomy; 23 agentic formulas, 5 proved in Lean.
v22	2026-06-03	Convergence: A5 merge, VCG proven on-branch, partial Cauchy closure, SLSA L2.
v23	2026-06-06	Unified Substrate: conditional Λ -uniqueness under declared A6'; unconditional uniqueness machine-checked FALSE; Λ stays Conjecture 1.

The line now reads as an unbroken lineage: v18 expands, v19 consolidates and verifies, v20 presents the verified anatomy, v21 ships the runtime, v22 converges, v23 unifies. v19 is the hinge on which expansion turns into verified anatomy.

12. Limitations, caveats, and open problems

- 1 No new proven mathematics.** v19 closes no sorry that was not already closed; its contribution is accounting and scope discipline, not new theorems.
- 2 The expansion track is mostly open.** Eighteen of nineteen 2026-05-30 modules carry open obligations (§4.2); v19 records this, it does not fix it.

- 3 **Λ remains Conjecture 1.** v19 preserves the label; it neither proves nor refutes uniqueness (those are later results).
- 4 **Counts are corpus-index counts.** The per-file counts are read from the corpus index, which was harvested with disclosed limits; the authoritative numbers are the locked kernel's (749, 14, 163) at c7c0ba17.
- 5 **SLSA L1+L2, not L3.** Supply-chain attestation is L1+L2; isolated, hardened builders (L3) are not claimed.

12.1 Threats to the honesty doctrine, and how the bridge mitigates them

A doctrine is only as good as its resistance to the pressures that erode it. The bridge anticipates four failure modes and answers each with a mechanical check rather than a promise:

Threat	How it would erode honesty	Bridge mitigation
Count inflation	folding experimental theorems into a headline 'proven' count	drift gate + experimental-scope exclusion; locked count pinned at 749/14/163
Silent sorry deletion	deleting an open obligation to present a green corpus	obligations are typed/named; deletion is a reviewable diff, not a cleanup
Axiom laundering	hiding an idealization inside a lemma so it disappears from the headline	#print axioms ledger surfaces every declared axiom per theorem
Conjecture promotion	quietly relabelling Λ from Conjecture 1 to theorem	counterexample is CI-green; the conjecture label is machine-caused (Nozick sensitivity)

Each mitigation is executable: none relies on a reviewer remembering to be honest. This is the deeper sense in which v19 is a *bridge* — it carries the honesty doctrine across from a stated intention (v14) to an enforced acceptance criterion (v20 onward), so that later versions inherit honesty as a CI invariant rather than as a cultural habit.

12.2 Open problems the bridge hands forward

- Close the highest-value expansion obligations (VCG truthfulness — later done on-branch in v22; Reed-Solomon Singleton bound; PAC-Bayes bounds).
- Discharge Λ -uniqueness via the n-dimensional Cauchy chain under A5 (partially closed in v22; still open on main).
- Promote the drift gate from a count-comparison to a per-declaration status diff, so that *which* theorem changed status is reported, not merely that a count moved.

13. Conclusion

The Verification Bridge is the version that turns a pile of modules into an auditable substrate. Its deliverable is a per-theorem verified index over the v18 expansion, honest about exactly which results are locked, which are sorry-free fragments, which are axiom-gated, and which are open; and a mechanical scope-separation discipline (the drift gate) that lets the substrate grow without contaminating its locked claims. That discipline is what makes the v20 "Culmination" a *verified* anatomy rather than an asserted one, and it is the same externalized meta-level the v23 unification later names explicitly.

One-line thesis. v19 is the bridge from expansion to verified anatomy: it replaces “how many modules” with “what does each theorem prove, and under what status,” and enforces the separation between a locked kernel of five formulas and an honestly-open experimental corpus.

14. Appendix A: Reproducibility and pins

The artifacts behind every claim, pinned for reproduction.

Artifact	Pin / identifier
Locked kernel commit	lutar-lean @ c7c0ba17
Locked counts	749 declarations / 14 unique axioms / 163 sorries (112 baseline + 51 Putnam)
Locked formulas	{F1, F11, F12, F18, F19}
Lean toolchain	Lean 4.13.0 (commit 6d22e0e5cc5a)
Concept DOI (always-latest)	10.5281/zenodo.19944926
v11 metrics DOI	10.5281/zenodo.20119582
ORCID	0009-0001-0110-4173
Drift-gate command	lake build then compare counts vs pinned baseline

15. Appendix B: Axiom ledger and notation glossary

The trusted base, stated once. The bottom turtles are: (i) the soundness of the Lean 4 kernel and its core axioms (propext, Quot.sound, Classical.choice); (ii) declared cryptographic idealizations (the SHA-256 collision-resistance axioms used by axiom-gated modules); (iii) the single declared governance axiom A6'_block_consistent, used only in the later conditional $\wedge$ theorem; and (iv) the human sign-off gating CI. Everything else is derived.

Symbol / term	Meaning
$\Lambda_k(x)$	Lutar invariant: equal-weight geometric mean $(\prod_i x_i)^{1/k}$.
sorry-free	A Lean development with no sorry placeholder; the kernel checks every step.
axiom-gated	Sorry-free given an explicitly declared, disclosed idealizing axiom.
#print axioms	Lean command listing the trusted axiom base a theorem depends on.
drift gate	CI invariant: locked counts (749/14/163) do not move when experimental modules are added.
Conjecture 1	The open claim that A6'-type block-consistency is the correct governance demand; never a theorem.
locked / experimental	Locked = pinned, kernel-checked scope; experimental = excluded until re-audited.

16. References

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Honesty doctrine preserved verbatim: Λ is Conjecture 1 unconditionally (never a theorem); the conditional uniqueness theorem holds only under the declared `A6'_block_consistent`; locked/proven = 5 {F1,F11,F12,F18,F19} @ c7c0ba17 (749/14/163); declared axioms disclosed in every `#print axioms` ledger; SLSA L1+L2 (not L3). No fabricated results, no fake citations.